CoPL: Collaborative Preference Learning for Personalizing LLMs

Youngbin Choi[1], Seunghyuk Cho[1], Minjong Lee [2], MoonJeong Park[1], Yesong Ko[2], Jungseul Ok[1,2], Dongwoo Kim[1,2,*]
[1]Graduate School of Artificial Intelligence, POSTECH,
[2]Department of Computer Science and Engineering, POSTECH,
{choi.youngbin, shhj1998, minjong.lee, mjeongp, yesong.ko, jungseul, dongwoo.kim}@postech.ac.kr

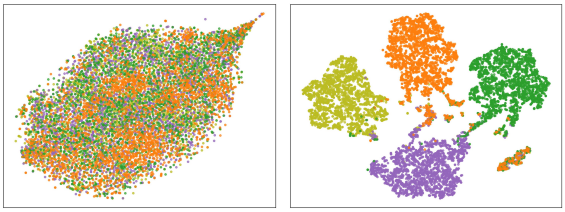
arXiv:2503.01658v2 [cs.LG] 17 Sep 2025

Abstract

Personalizing large language models (LLMs) is important for aligning outputs with diverse user preferences, yet existing methods struggle with flexibility and generalization. We propose CoPL (Collaborative Preference Learning), a graph-based collaborative filtering framework that models user-response relationships to enhance preference estimation, particularly in sparse annotation settings. By integrating a mixture of LoRA experts, CoPL efficiently fine-tunes LLMs while dynamically balancing shared and user-specific preferences. Additionally, an optimization-free adaptation strategy enables generalization to unseen users without fine-tuning. Experiments on TL;DR, UltraFeedback-P, and PersonalLLM datasets demonstrate that CoPL outperforms existing personalized reward models, effectively capturing both common and controversial preferences, making it a scalable solution for personalized LLM alignment. The code is available at `https://github.com/ml-postech/CoPL`.

1 Introduction

Large language models (LLMs) have rapidly expanded across diverse applications, from customer service and tutoring to creative content generation (Shi et al., 2024; Molina et al., 2024; Venkatraman et al., 2024). As increasing numbers of users with varied backgrounds interact with LLMs, accounting for diverse preferences has become essential. Most reward models rely on the Bradley-Terry-Luce (BTL) framework (Bradley and Terry, 1952), which learns preferences from pairwise comparisons provided by human annotators. However, earlier studies largely assumed a single, uniform preference and neglected the diversity of user preferences (Siththaranjan et al., 2024; Li et al., 2024, 2025). This limitation has led to growing interest in personalized reward models (Sorensen et al., 2024; Liu et al., 2025; Guan et al., 2025).



(a) VPL (b) CoPL

Figure 1: T-SNE visualization of seen user embeddings in UF-P-4 (AVG) with `gemma-2b-it`. Points are colored by their preference group. Our method clusters users in the same group more effectively. T-SNE visualizations of other baselines are provided in Fig. A1.

There are two different approaches to utilizing the BTL framework for personalized reward models. The first approach has explored combining multiple reward models, each trained for a specific preference and later aggregated (Jang et al., 2023; Oh et al., 2024). However, this approach relies on pre-trained models for different preference types, reducing flexibility. Another line of work introduces user-specific latent variables into a single BTL framework, learning personalized representations from user annotations (Chen et al., 2024a; Poddar et al., 2024; Li et al., 2024; Barreto et al., 2025). While this method captures individual preferences, the latent variable model does not explicitly account for relationships between users sharing similar responses. As a result, it struggles to generalize in sparse annotation settings.

To address these limitations, we propose Collaborative Preference Learning (CoPL), which constructs a user-response bipartite preference graph from pairwise annotations and uses a graph-based collaborative filtering (GCF) framework for personalized reward modeling. Unlike approaches that model each user separately, GCF on the graph structure allows preference signals to propagate across

*Correspondence to: Dongwoo Kim <dongwoo.kim@postech.ac.kr>

users, enabling to exploit multi-hop relationships among users and responses (Wang et al., 2019; He et al., 2020). CoPL can capture diverse preferences of users even in sparse annotation settings.

When annotations are sparse, latent-variable methods face significant challenges, as the scarcity of supervisory signals makes it difficult for randomly initialized user representation encoders to converge toward semantically meaningful representations. As a result, users with similar underlying preferences can sometimes be mapped to distant points in the latent space if their annotated response pair sets do not overlap. In such cases, sparse supervision may cause semantically similar users to appear unrelated in the learned embedding space. For instance, consider three users: user 1 annotates the pairs $(a, b), (c, d)$, user 2 annotates $(c, d), (e, f)$, and user 3 annotates $(e, f), (g, h)$ with the same preference. Although user 1 and user 3 exhibit similar preferences, the lack of overlapping annotations provides no direct signal for aligning their representations. CoPL addresses this issue by constructing a user–response bipartite graph and propagating preference signals through multi-hop message passing. This mechanism enables the alignment of users with disjoint annotation sets, such as user 1 and user 3, thereby providing better data efficiency and generalization. Fig. 1 illustrates that, under sparse annotation, CoPL produces embedding spaces in which users with identical preferences are more coherently aligned.

Based on the user embedding, we develop an LLM-based reward model that can predict the preference score of a user given input text. We adopt the mixture of LoRA experts (MoLE) (Chen et al., 2023, 2024c; Liu et al., 2024) that allows parameter-efficient fine-tuning while routing different users to different paths based on the learned embedding. Specifically, we develop a user preference-aware gating function that dynamically selects the experts in the forward pass, making the LLM predict a personalized preference.

While the reward model can predict preferences for users included in the training set, the model cannot handle newly participated *unseen* users whose embeddings are unknown. To estimate the preferences of unseen users, we propose an optimization-free adaptation method. Given a few annotations from an unseen user, we exploit the existing graph to find users with similar preferences and aggregate their embeddings to represent the unseen user.

Experimental results demonstrate that CoPL consistently outperforms existing personalized reward models in both seen and unseen users. Especially, CoPL generalizes to unseen users, maintaining high accuracy with only a few provided annotations. Embedding visualizations show that CoPL clusters users with similar preferences more closely than competing baselines. Further ablation studies confirm that both GCF and MoLE contribute significantly to performance.

2 Related Work

Alignment has emerged as a crucial strategy for mitigating undesirable outcomes (Dai et al., 2023; Yang et al., 2024a). Previous research has often focused on the average preference of annotators (Achiam et al., 2023), ignoring the diverse preferences. To address preference diversity, recent works (Jang et al., 2023; Oh et al., 2024; Yang et al., 2024b) view this problem as a soft clustering problem, where user-specific preferences are treated as mixtures of predefined preference types. Although this approach effectively handles diverse preferences, it relies on specifying several preference types in advance.

Another line of work introduces a user latent variable in the BTL framework (Poddar et al., 2024; Li et al., 2024; Chen et al., 2024a). The main challenge lies in obtaining user representations. One approach is to treat each user embedding as learnable parameters (Li et al., 2024; Chen et al., 2024a), and the other strategy is to train an encoder that infers embeddings from the set of annotated pairs provided by each user (Poddar et al., 2024).

We also discuss preference learning with sparse interactions, closely related to our approach, in Section C.

3 Problem Formulation

We aim to develop a reward model that can capture diverse user preferences from a limited set of preference annotations. Instead of directly defining a user's preference, we collect pairwise comparisons indicating which item a user prefers. Let $\mathcal{U} = \{1, \cdots, U\}$ be a set of users and $\mathcal{X}$ be a space of LLM's responses. To estimate the preferences of users, we first curate a *survey set* $\mathcal{S} = \{(q_i, a_i, b_i)\}_{i=1}^{R}$ consisting of predefined questions q_i and two different responses $a_i, b_i \in \mathcal{X}$ from LLMs. For each user u, we first randomly sample N_u number of survey items and then collect the preferences over the response pairs, resulting

in *preference dataset* $\mathcal{D}_u$. We use $(a \succ b) \in \mathcal{D}_u$ to denote that user u prefers response a over the response b. Given these pairwise preferences, we aim to learn a numerical reward function

$$f(u, r) : \mathcal{U} \times \mathcal{X} \to \mathbb{R}, \tag{1}$$

where $f(u, r)$ represents a scalar *preference score* of response r for user u. The model is trained to satisfy

$$f(u, a) > f(u, b)$$

for all u and preference pairs $a \succ b$ observed in the data.

Following previous works (Li et al., 2024; Poddar et al., 2024), we consider the Bradly-Terry-Luce (BTL) choice model (Bradley and Terry, 1952) with maximum likelihood estimation to train the reward function. The likelihood of user u prefers item a over b can be defined using the BTL model as

$$p(a \succ b \mid u) = \frac{\exp\big(f(u, a)\big)}{\exp\big(f(u, a)\big) + \exp\big(f(u, b)\big)}.$$

Conversely, if b was chosen over a, i.e., $a \prec b$, the likelihood is

$$p(b \succ a \mid u) = 1 - p(a \succ b \mid u).$$

Through the maximum likelihood estimation with preference data for all users, one can learn the reward function f to make the reward function align with user preference. In the case of the universal preference model, user u is ignored in Eq. (1) (Chen et al., 2024b; Achiam et al., 2023; Dai et al., 2023; Bai et al., 2022). In practice, the user u is replaced by a user embedding (Poddar et al., 2024; Li et al., 2024; Chen et al., 2024a).

4 Method

In this section, we describe our Collaborative Preference Learning (CoPL). Our approach consists of three steps: learning user representations given preference data, construction of personalized reward models, and adaptation to unseen (new) users at test time. Figure 2 illustrates the first two steps, and Figure 3 the last step.

4.1 User Representation Learning

Users who share similar preferences are likely to respond to similar responses. When the number of annotated responses is very small, it is unlikely to annotate the same responses between users. However, if we exploit multi-hop relations between users and responses, we may estimate user preference accurately. In fact, the exploitation of the relationship between users and items is the key idea behind graph-based collaborative filtering (GCF).

The preference dataset for all users can be naturally converted into a bipartite graph, where each user and response is represented as a node, and an edge between a user and a response represents the user's preference over the response, as illustrated in Fig. 2. The edge can have two different types: positive or negative, indicating whether a user prefers the response or not.

Given a bipartite graph, we design a message-passing algorithm to update user and response representations. Let $\boldsymbol{e}_u \in \mathbb{R}^d$ be an embedding vector of user u, and $\boldsymbol{e}_r \in \mathbb{R}^d$ be an embedding vector of response r. Since there are two different edge types, we use different parameterizations for each type. Let $\mathcal{N}_u^+$ be a set of positive edges and $\mathcal{N}_u^-$ be a set of negative edges from user u. Similary, we can define $\mathcal{N}_r^+$ and $\mathcal{N}_r^-$ for response r. Given user and response embeddings at layer ℓ, the message passing computes a message from neighborhood responses to the user as

$$\begin{aligned}
\boldsymbol{m}_u^+ &= \sum_{r \in \mathcal{N}_u^+} \alpha_{u,r} \Big(W_1^{(\ell)} \boldsymbol{e}_r^{(\ell)} + W_2^{(\ell)} (\boldsymbol{e}_r^{(\ell)} \odot \boldsymbol{e}_u^{(\ell)}) \Big), \\
\boldsymbol{m}_u^- &= \sum_{r \in \mathcal{N}_u^-} \beta_{u,r} \Big(W_3^{(\ell)} \boldsymbol{e}_r^{(\ell)} + W_4^{(\ell)} (\boldsymbol{e}_r^{(\ell)} \odot \boldsymbol{e}_u^{(\ell)}) \Big), \\
\boldsymbol{m}_u^{(\ell)} &= W_{\text{self}}^{(\ell)} \boldsymbol{e}_u^{(\ell)} \; + \; \boldsymbol{m}_u^+ \; + \; \boldsymbol{m}_u^-,
\end{aligned} \tag{2}$$

where $W_1^{(\ell)}, W_2^{(\ell)}, W_3^{(\ell)}, W_4^{(\ell)}, W_{\text{self}}^{(\ell)} \in \mathbb{R}^{d \times d}$ are parameter matrices, $\odot$ is element-wise multiplication, and $\alpha_{u,r}$ and $\beta_{u,r}$ are normalization factors, set to $\frac{1}{\sqrt{|\mathcal{N}_u^+||\mathcal{N}_r^+|}}$ and $\frac{1}{\sqrt{|\mathcal{N}_u^-||\mathcal{N}_r^-|}}$, respectively. Then, the user embedding is updated with the aggregated message $\boldsymbol{m}_u^{(\ell)}$:

$$\boldsymbol{e}_u^{(\ell+1)} = \psi\big(\boldsymbol{m}_u^{(\ell)}\big), \tag{3}$$

where $\psi(\cdot)$ is a non-linear activation. The response embedding $\boldsymbol{e}_r^{(\ell)}$ is updated with analogous process. We randomly initialize the user and response embeddings at the first layer and then fine-tune the embeddings through training. The update steps for the response embeddings are provided in Section A.